

MinHash

Estimating Jaccard Similarity at Scale

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Finding Similar Documents

window-4 shingling: "A brown cat sat on the mat"

→ {"A br", " bro", "brow", "rown", "own ", "wn c", ...}

Treat each document as a set of shingles.

Compare the sets for similarity.

Finding Similar Documents

Goal: Find near-duplicate documents in a large corpus.

- ▶ Web pages reposted with minor changes.
- ▶ Plagiarism detection.
- ▶ Product deduplication.

Naive: Compare all pairs $\Rightarrow O(n^2)$ comparisons. Each comparison is also slow.

Jaccard Similarity

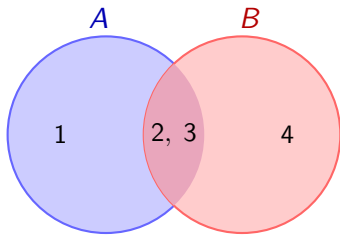
Represent a document as a **set of shingles** (words or character n -grams).

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

$J = 0$	$J = 0.5$	$J = 1$
disjoint	partial overlap	identical

Example

$$A = \{1, 2, 3\}, \quad B = \{2, 3, 4\}$$



$$J(A, B) = \frac{|\{2, 3\}|}{|\{1, 2, 3, 4\}|} = \frac{2}{4} = 0.5$$

Characteristic Matrix

One row per universe element, one column per set.

element	A	B
1	1	0
2	1	1
3	1	1
4	0	1

Row type **(1,1)**: element in $A \cap B$ (rows 2, 3)

Row type **(1,0) or (0,1)**: element in one set only (rows 1, 4)

Permutation Concept

Randomly permute the rows. Define:

$\text{minhash}(S)$ = first element of S under the permutation

Key: the very first row in $A \cup B$ is type (1,1) iff $\text{minhash}(A) = \text{minhash}(B)$.

$$\Pr[\text{minhash}(A) = \text{minhash}(B)] = \frac{|A \cap B|}{|A \cup B|} = J(A, B)$$

Why?

Consider the first row in $A \cup B$ under a random permutation.

- ▶ It is a **uniformly random** row among all rows in $A \cup B$.
- ▶ It lands in $A \cap B$ (type 1,1) with probability $\frac{|A \cap B|}{|A \cup B|} = J(A, B)$.
- ▶ That is exactly the event $\text{minhash}(A) = \text{minhash}(B)$.

Permutation: No Match

Permutation order: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

order	A	B
1	1	0
2	1	1
3	1	1
4	0	1

First in $A \cup B$: element **1** (type 1, 0).

$\text{minhash}(A) = 1 \neq 2 = \text{minhash}(B) \Rightarrow \text{no match}$

Permutation: Match

Permutation order: $2 \rightarrow 1 \rightarrow 4 \rightarrow 3$

order	A	B
2	1	1
1	1	0
4	0	1
3	1	1

First in $A \cup B$: element **2** (type 1, 1).

$\text{minhash}(A) = 2 = \text{minhash}(B) \Rightarrow \text{match!}$

From Permutations to Hash Functions

Problem: Storing a true random permutation of $|U|$ elements needs $O(|U|)$ space.

Solution: Use a hash function to *simulate* a permutation.

MinHash via hashing

$$\text{minhash}_h(S) = \arg \min_{x \in S} h(x)$$

Each distinct hash function $\approx$ one random permutation of the universe.

Universal Hash Family

Choose prime $p > |U|$. Pick random $a \in [1, p)$, $b \in [0, p)$.

$$h_{a,b}(x) = (ax + b) \bmod p$$

- ▶ Different (a, b) pairs give independent “permutations”.
- ▶ For any $x \neq y$: the pair $(h(x), h(y))$ is uniformly random over distinct values.

MinHash Sketch

Input: Set S , k hash functions $h_1, \dots, h_k$.

Sketch:

$$\text{sig}(S) = \left(\min_{x \in S} h_1(x), \min_{x \in S} h_2(x), \dots, \min_{x \in S} h_k(x) \right)$$

Estimate:

$$\hat{J}(A, B) = \frac{1}{k} \# \left\{ i : \min_{x \in A} h_i(x) = \min_{x \in B} h_i(x) \right\}$$

Hash Values: $p = 5$

$$A = \{1, 2, 3\}, \quad B = \{2, 3, 4\}, \quad J(A, B) = \frac{1}{2}$$

$h_{a,b}(x) \bmod 5$	$h(1)$	$h(2)$	$h(3)$	$h(4)$
$h_1 : x$	1	2	3	4
$h_2 : 3x + 1$	4	2	0	3
$h_3 : 2x + 1$	3	0	2	4
$h_4 : 4x$	4	3	2	1

Signatures & Estimation

	$\min_A$	$\min_B$	match?
$h_1 : x$	1	2	no
$h_2 : 3x + 1$	0	0	yes
$h_3 : 2x + 1$	0	0	yes
$h_4 : 4x$	2	1	no

$$\hat{J}(A, B) = \frac{2}{4} = 0.5 \quad (\text{true } J = 0.5)$$

Unbiased Estimator

Each indicator $X_i = \mathbf{1}[\text{minhash}_i(A) = \text{minhash}_i(B)]$ is Bernoulli(J).

$$\mathbb{E}[\hat{J}] = \mathbb{E}[X_i] = J$$

$\hat{J}$ is an **unbiased** estimator of J for any $k \geq 1$.

Error Bound

$$\text{Var}[\hat{J}] = \frac{J(1-J)}{k} \leq \frac{1}{4k}$$

$$\sigma = \sqrt{\frac{J(1-J)}{k}} \leq \frac{1}{2\sqrt{k}}$$

k	$\max \sigma$
25	0.10
100	0.05
400	0.025

Space and Time

- ▶ **Sketch:** k integers per set.
- ▶ **Build:** $O(k \cdot |S|)$ hash evaluations.
- ▶ **Compare:** $O(k)$ integer comparisons.
- ▶ **Original set:** not needed after sketching.

Summary

- ▶ MinHash estimates Jaccard similarity using only k integers per set.
- ▶ Key property: $\Pr[\text{minhash}(A) = \text{minhash}(B)] = J(A, B)$.
- ▶ Accuracy: $\sigma \leq \frac{1}{2\sqrt{k}}$; use $k = 100\text{--}400$ in practice.
- ▶ **Applications:** near-duplicate detection, clustering, recommendation.